MIL-L-83176

23 September 1969

MILITARY SPECIFICATION

LUBRICANT, INSTRUMENT BEARING, PETROLEUM BASE

This specification is mandatory for use by all Departments and agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for specially refined paraffinic petroleum base lubricant.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

P-D-680	Dry Cleaning Solvent
DD-B-00585	Bottles and Jars, Prescription, Dispensing
DD-B-606	Containers, One (1) Gallon
QQ-C-576	Copper Flat Products with Slit, Slit and Edge Rolled, Sheared, Sawed, or Machined Edges (Plate, Bar, Sheet, and Strip)
QQ-S-698	Steel, Sheet and Strip, Low Carbon
TT-T-656	Tricresyl Phosphate

STANDARDS

Federal

Fed Test Method Std No. 791	Lubricants; Liquid Fuels, and Related Products; Methods of Testing

FSC 9150

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Military

MIL-STD-105	Sampling Procedures and Tables For Inspection by Attributes
MIL-STD-290	Packaging, Packing and Marking of Petroleum and Related Products

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials Publication

ASTM Standards on Petroleum Products and Lubricants, Parts 17 and 18.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

3.1 Preproduction. The lubricant furnished under this specification shall be a product which has been tested and passed the preproduction tests specified herein on the first processed lot.

3.2 Materials. The materials used in compounding the lubricant shall consist of and be limited to a natural paraffinic base stock derived from Pennsylvania Crude Oil and shall contain additives as specified. The base stock shall be a molecularly distilled product obtained by short path distillation and finished to color by filtration using a bauxite type absorbent. No re-refined products shall be used.

3.2.1 Additives. Additives other than specified herein shall not be used.

3.2.1.1 Oxidation inhibitor. The oxidation inhibitor shall be a hindered bis-phenol. (An example of a known suitable additive is Ethyl 702 4,4' methylenebis (2,6-di-tertiary butyl phenol manufactured by the Ethyl Corporation at 98 mole percent purity (dry basis)). A quantity of 0.45 to 0.55 percent by weight shall be used.

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3.2.1.2 Antiwear agent. The antiwear agent shall be tricresyl phosphate conforming to TT-T-656. A quantity of 0.90 to 1.10 percent by weight shall be used in the lubricant.

3.3 Properties of the lubricant. The properties of the lubricant shall conform to the requirements of table I and 3.3.1 through 3.3.4.

Table I. Properties of Lubricant

PROPERTY	VALUE
Gravity (API)	Report
Flash Point, Min., °F	500
Viscosity 100°F, Kinematic, CS	Report
Viscosity 210°F, Sus , Kinematic, CS	77.01 - 81.0 14.76 - 15.78
Viscosity Index, Min.	100
Pour Point, °F Max.	25
Sulphur, Wt % Max.	0.10
Color, ASTM Max.	1/2
Trace Sediment, Volume %, Max.	0.025

3.3.1 Gravity. The gravity of the lubricant shall be determined but not limited (see 4.5). Samples of lubricant submitted for acceptance tests shall not vary more than ±0.006 at 60°F/60°F (15.6°C/15.6°C) from the gravity of the sample approved by preproduction testing.

3.3.2 Corrosiveness and oxidation stability. When tested as specified in 4.5.1, the change in weight of the steel specimen shall be not more than 0.2 milligram (mg) per square centimeter of surface, and the weight change in the copper shall be not more than 0.6 mg per square centimeter of surface.

3.3.2.1 Resistance to oxidation. The lubricant viscosity shall be within -5 to +20 percent of the original viscosity in centistokes at 100°F (37.8°C) after the corrosiveness and oxidation stability test (see 4.5.1). After oxidation, the total acid or base number shall be not more than 2.0.

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3.3.3 Lubricity. When tested in accordance with 4.5.2, wear values for the lubricant shall be as specified in table II.

Table II. Wear values

Load (kg)	Average Wear Scar diameter mm (max)
1	0.20
10	0.30
40	0.65

3.3.4 Compatibility. The lubricant shall be compatible in all concentrations with each of the fluids qualified under this specification when tested as specified in 4.5.3.

3.3.5 Workmanship. After final additive compounding, the product shall be passed through a nominal 5-micron filter (25 micron absolute) into containers that are clean and dry. This finished filtered product shall be homogenous, and free of undissolved water and sediment at room temperature (70°F to 80°F). Additive solubility shall be completed so as not to be depleted by subsequent operational filtration through 0.45-micron filter elements in bearing applications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemend necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection and testing of the lubricant shall be classified as preproduction tests (first processed lot) and quality conformance tests.

4.3 Preproduction testing

4.3.1 Preproduction test samples. Preproduction test samples shall consist of 1 gallon of the lubricant, 8 ounces (fluid) of tricresyl phosphate, and 1 ounce of the oxidation inhibitor. In the event that additives are supplied as concentrated solutions, an equivalent quantity of solution shall be furnished. Samples shall be prominently identified by securely attached, durable tags marked with the manufacturer's own number and any additional information required by the letter of authorization (see 6.3). The preproduction inspection shall consist of

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testing the first processed lot sample for section 3 requirements deemed necessary by the testing activity designated (see 6.2). Delivery of the lubricant will be deferred pending completion of the first processed lot inspection. If the results of the first processed lot inspection are in accordance with the requirements of this specification, the consistency of production quality will be thereby verified. Failure of the lubricant to pass a first processed lot inspection shall require that the acceptance and further shipment of the lubricant be withheld until the contractor has corrected the conditions which led to the failure. Cleanliness inspection shall be conducted immediately prior to packaging the product.

4.3.1.1 Test report. Preproduction samples shall be accompanied by a certified test report containing complete information as to the source and type of base stock and additive materials used, the formulation and composition of the lubricant, and laboratory data showing quantitative results of all tests required by this specification. Any change of the source and type of base stock and additive materials used, the formulation and composition shall be cause for the submittal of a new preproduction sample by the supplier.

4.3.1.1.1 Formulation sheets. An example of a satisfactory form for the formulation sheet, indicating the percentage and nature of each ingredient, is as follows:

Hydrocarbon base stock (composition)	weight percent
Tricresyl phosphate additive (mfr's name and no.)	weight percent
Oxidation inhibitor (mfr's name and no.)	weight percent

4.3.2 Preproduction tests. Preproduction tests shall consist of tests for all requirements specified in section 3.

4.4 Quality conformance inspections

4.4.1 Inspection requirements. Inspection shall be in accordance with method 9601 of Fed Test Method Std No. 791.

4.4.2 Sampling for lot acceptance

4.4.2.1 Lot. A lot shall consist of all lubricant manufactured from one base stock as one batch and offered for delivery at one time. A batch is defined as the end product of all raw materials mixed or blended in a single operation.

4.4.2.2 Samples for quality conformance tests. Samples for quality conformance tests (see 4.4.5) shall be selected from each lot in accordance with method 8001 of Fed Test Method Std No. 791 (equivalent to ASTM Standard D270).

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4.4.2.3 Samples for inspection of filled containers. A random sample of filled containers shall be selected from each lot in accordance with MIL-STD-105 at inspection level II, and acceptance quality level (AQL) of 2.5 percent defective to verify compliance.

4.4.3 Inspection of empty containers. Prior to filling, each empty unit container shall be visually inspected for cleanliness and suitability.

4.4.4 Inspection of filled containers and preparation for delivery requirements. Each container selected in accordance with 4.4.2.3 shall be visually examined for defects of construction of the container and the closure, for evidence of leakage, for markings, and all other defects.

4.4.5 Quality conformance tests. Quality conformance tests for acceptance of individual lots shall consist of tests for the following requirements specified in section 3:

a. Viscosity at 210°F (98.9°C).

b. Viscosity Index

c. Flash point

d. Trace sediment

e. Sulphur

f. Color

4.5 Test methods. Tests shall be conducted in accordance with Fed Test Method Std No. 791, or ASTM Standards, using applicable method as listed in table III, and 4.5.1 through 4.5.3.

Table III. Test Methods

	Test Method	
	Fed Std No. 791	ASTM Standard
Pour point	201	D97
Flash Point	1103	D92
Trace Sediment	3004	—
Gravity	401	D287
Viscosity	305	D445
Viscosity Index	9111	D2270
Color,	102	D1500
Sulphur	—	D1551
Viscosity, conversion of kinematic to Saybolt	—	D2161

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4.5.1 Corrosiveness and oxidation stability. The lubricant shall be tested in accordance with method 5308 of Fed Test Method Std No. 791 except as specified herein. The lubricant sample shall be 100 ml. The temperature of the bath shall be 300°F. The test metals shall be copper conforming to QQ-C-576 and steel, carbon, mild conforming to QQ-S-698. The test shall be run for 48 hours at 5 ± 0.5 liters per hour flow rate of air.

4.5.2 Lubricity

4.5.2.1 Apparatus. The apparatus for the lubricity test shall consist of a Shell Four-Ball Wear Tester (see 6.4), an Alnor temperature Controller or equivalent, 1/2-inch AISI-C-52100 steel balls (SKF Grade 25-EP or equivalent) for use with the Four-Ball Wear Tester, and wrenches supplied with the tester.

4.5.2.2 Preparation. The steel balls, chuck, and ball pot assembly shall be individually and thoroughly washed in solvent conforming to P-D-680, rinsed in clean solvent and in petroleum ether, then dried by allowing the solvent to evaporate. Four clean balls shall be selected. One ball shall be placed in the revolving chuck ball holder. The chuck shall be firmly tightened and placed in the machine. Three balls shall be placed in the ball pot assembly and locked in by tightening the cup sufficiently to prevent any movement or rotation of the balls during the test. The secured balls in the ball pot assembly shall be covered with an ample covering of the test fluid and the ball pot assembly placed in position in the machine.

4.5.2.3 Test procedure. The speed of the tester shall be adjusted to 600 ± 20 rpm and run for one (1) hour at each load specified in table II. The mercury cup assembly shall be moved up to make contact with the heater and thermo-couple terminals of the ball pot assembly. The temperature controller shall be adjusted to 167°F ± 2°F (75°C ± 1°C), the heater switch turned on, and the variable transformer dial adjusted until the unit has lined out at the desired test temperature. The motor shall then be turned off after one hour, the mercury cup lowered, and the load released. The ball pot assembly shall be removed from the test machine, flushed thoroughly with petroleum ether, and dried with a clean, soft cloth. To facilitate locating the scars when the balls are viewed under a microscope a circle shall be inscribed with a diamond tip pencil around the scar area of each of the three balls. The balls shall then be removed from the ball pot assembly, cleaned in naphtha and petroleum ether, and placed in a suitable holder (a small wooden block) on a microscope stage. The minimum diameter of the scar and the diameter at 90° to this position shall be measured on each of the three balls. The average diameter, based on six readings, shall be calculated and reported in millimeters (mm).

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4.5.3 Compatibility. Lubricant in the volumes of 20 ml, 100 ml, and 180 ml shall be mixed with fluid from each manufactured source previously accepted under this specification and a total of 200 ml volumes made. These mixtures shall be prepared in 250-ml glass-stoppered flasks. The flash shall be thoroughly agitated and then stored in an oven at 210°F±2°F for 2 hours. At the end of this time none of the mixtures shall show any signs of sediment, turbidity, or crystallization. The samples shall then be stored at 100°F±2°F for a period of 2 hours. None of the mixtures shall show any signs of sediment, turbidity or crystallization.

4.6 Rejection. Failure of any sample of lubricant to conform to any of the requirements of this specification shall be cause for rejection of the lot represented.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. Packaging, packing, and marking of the lubricant shall be in accordance with MIL-STD-290. The type and size of the container shall be as specified by the procuring activity (see 6.2). The container material and the lubricant shall not have any adverse effect on each other. Prior to filling, containers shall be purged of dirt, water, or other contaminant which would degrade the quality of the lubricant.

5.1.1 Additional marking. In addition to any special marking required in the contract or order, the unit container shall be marked as follows:

> INSTRUCTIONS: DESTROY THIS CONTAINER WHEN EMPTY. DO NOT MIX WITH ANY FLUIDS EXCEPT THOSE OF MIL-L-83176 AND REVISIONS THERETO.
>
> WARNING
>
> CONTAINS TRICRESYL PHOSPHATE - PRODUCES PARALYSIS IF TAKEN INTERNALLY. DO NOT USE AS MEDICINAL OR FOOD PRODUCT OR IN FOOD MACHINERY ON SURFACES THAT MAY CONTACT FOOD. THE OIL MUST NOT BE ALLOWED TO CONTAMINATE FOODSTUFFS.

6.1 Intended use. The lubricant covered by this specification is intended for use in the spin axis bearings of inertial guidance gyros, accelerometers and other suitable instrument applications.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Type and size of container (see 5.1 and 6.2.1).

c. Quantity.

d. Selection of applicable levels of packaging and packing (see 5.1).

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6.2.1 Unit of purchase. The material should be purchased by volume, the unit being a U. S. Gallon at 60°F (15.6°C). For Air Force use, the lubricant will not be purchased in other than 1-quart or 1-gallon amber glass containers. The applicable container and screw cap specifications are as follows:

1 gallon - DD-B-606

1 quart - DD-B-00585, Type II, Style 1, amber, 32 fl. oz.

Screw caps and liners are to be plastic and polyethylene lined as specified by the above specification.

6.3 Preproduction tests. The activity responsible for receiving the preproduction samples and data is the Air Force Materials Laboratory. Attn: MAAE, Wright-Patterson AFB, Ohio 45433. Information pertaining to preproduction samples may be obtained from that activity.

6.4 Source. The Shell Four-Ball Wear Tester may be obtained under Catalog No. 73603, from the Precision Scientific Company, 3737 West Cortland, Chicago, Illinois, 60607.

CUSTODIANS:
Army - MR
Navy - AS
Air Force - 11

REVIEWER ACTIVITIES:
Army - MR, WC, ME, MU, MD
Navy -
Air Force - 68

PREPARING ACTIVITY:
Air Force - 11

Project Number - 9150-0258

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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